

Module 7

Financial Management

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

Strategic management consists of top-level management activities usually undertaken by the board of directors and senior management of a company. The board of directors of a company sets the mission, vision and goals of a company to be achieved in the long term, sometimes beyond five or ten years.

Based on these directions, the senior management of the company will formulate the company's strategic plans (e.g. marketing, sales, financial, etc.), which are implemented and quantified by annual budgets, used as financial targets for planning, control and performance measurement; as part of the management accounting processes.

Answer 1(b)

A business plan is a road map that outlines the results of careful strategic planning and provides a blue-print how a business will be run and grow over time.

A business plan describes in detail how a business will evolve. A well-crafted business plan is effective to help an entrepreneur in securing fund raising and communicating any expectations for what the business will look like and how it will get there, and the returns investors can expect.

In addition, a business plan can help entrepreneurs set goals and objectives, provide benchmarks for evaluating and controlling company performance, and communicate results to managers and other stakeholders such as customers and suppliers.

Answer 1(c)(i)

The depth and breadth of business plans vary widely but the following basic components could be included in the business plan for the Group:

1. Executive summary of the Group's mission or vision
 - The mission and vision of the Group is to create modern life aesthetics by providing one-stop medical and healthcare services to realise a "healthy and beautiful life".
2. Overview of the Group, with description of the products and services, and the marketing and sales activities for its target customers
 - As a leader in the industry, with passionate enthusiasm for medical cosmetology, the Group established its first flagship aesthetic medical centre at Causeway Bay in the early 2000s.

- The Group currently operates over 280 clinics/ centres, providing a full range of medical health and medical beauty services to both male and female customers of the high, mid-range and mass markets in the Greater Bay Area.

Answer 1(c)(ii)

Despite the fact that the business has taken a hard hit due to the pandemic, it is unlikely to shake the aspiration of consumers for beauty as indicated by the surge demand of nail care products ("2019 pandemic lipstick effect") and the increased eye-cosmetic products sales revenues generated in the United Kingdom, the United States and mainland China.

In addition, dermatologists and beauty professionals have highlighted the opportunities mandatory quarantines grant those interested in cosmetic surgery: the imperative to stay indoors for a period of time makes this an ideal time to engage a skincare treatment and/ or plastic surgery that might otherwise take days/ weeks for recuperation in secrecy.

After a prolonged period of time being home office, many people may tend to gain weight quickly because of inactivity. As a result, it is not surprising to anticipate that there will be a pressing and sustainable demand for the new all-rounded medical health and beauty services in the areas of skincare and plastic surgery, together with a tailor-made keep-fit program, after the pandemic.

(Any sensible descriptions or explanations are acceptable)

Answer 1(d)

Present value ("PV") of first annual cash flow: $\frac{\text{HK\$}950,000}{(1+0.12)^3} = \text{HK\$}676,191.24$

PV of subsequent annual cash flows that will grow at 5% perpetually:

$$\frac{\frac{\text{HK\$}950,000 \times 1.05}{0.12 - 0.05}}{(1+0.12)^3} = \text{HK\$}10,142,868.53$$

PV of the new project given the discount rate is 12%:

$$\text{HK\$}676,191.24 + \text{HK\$}10,142,868.53 = \text{HK\$}10,819,059.77$$

Answer 2(a)

Peter's estimation is incorrect.

The amount calculated by Peter is actually referring to the current yield of the bonds, but not the cost of debt. The cost of debt should be the yield to maturity ("YTM") of the bonds that includes both current yield and capital gain yield.

The calculation of the current yield includes only the cost of bond interest payments but ignores the fact that Webster has to repay the par value (HK\$1,000) of the bond upon maturity.

The difference between the market price of HK\$860 and par value of HK\$1,000 represents an additional financing cost to Webster.

Besides, as the interest expense of the bond is tax-deductible, the actual cost of debt should be after-tax, which is lower than what it is.

Answer 2(b)

Peter's conclusion is incorrect.

The cost of internal equity consists of both dividend yield and capital gain yield.

Peter's calculation ignores the capital gain yield component and therefore understates the cost of internal equity.

Moreover, cost of internal equity is not the same as that of external equity through share issuance on the public stock exchange. External equity incurred additional flotation cost which results in higher financing costs than internal equity.

Answer 2(c)

One-year interest = $r = 10\%$

Two-year effective interest = $R = (1 + 10\%)^2 - 1 = 21\%$

PV_0 of the HK\$4 cash flow stream = $\frac{HK\$4}{R} = \frac{HK\$4}{0.21} = HK\$19.05$

The market price per share is HK\$19.05.

Answer 3(a)

		Project RR HK\$	Project AD HK\$
Total cash inflows	CF1 to CF4 / 5	1,354,400	1,048,000
Less: Depreciation	(Cost x 90%)	(873,000)	(616,500)
Total profits		<u>481,400</u>	<u>431,500</u>
Average profit	Total profits / 4 or 5 years	120,350	86,300
Average investment	(Beg + End) / 2	533,500	376,750
Accounting rate of return		<u>22.56%</u>	<u>22.91%</u>

Project AD should be accepted due to higher ARR.

Answer 3(b)

Net present value ("NPV") of Project RR:

$$\begin{aligned}\text{NPV(RR)} &= - \text{HK\$}970,000 + \frac{\text{HK\$}450,900}{1.08} + \frac{\text{HK\$}387,000}{1.08^2} + \frac{\text{HK\$}220,500}{1.08^3} + \frac{\text{HK\$}296,000}{1.08^4} \\ &= \text{HK\$}171,898.97\end{aligned}$$

NPV of Project AD:

$$\begin{aligned}\text{NPV(AD)} &= - \text{HK\$}685,000 + \frac{\text{HK\$}225,000}{1.08} + \frac{\text{HK\$}257,000}{1.08^2} + \frac{\text{HK\$}275,000}{1.08^3} + \frac{\text{HK\$}190,500}{1.08^4} + \frac{\text{HK\$}100,500}{1.08^5} \\ &= \text{HK\$}170,395.07\end{aligned}$$

Project RR should be accepted due to higher NPV.

Answer 3(c)

Payback period of Project RR:

$$2 \text{ years} + \text{HK\$}132,100 / \text{HK\$}220,500 = 2.60 \text{ years}$$

Payback period of Project AD:

$$2 \text{ years} + \text{HK\$}203,000 / \text{HK\$}275,000 = 2.74 \text{ years}$$

Project RR should be accepted due to having shorter payback period.

Answer 3(d)

Accounting rate of return has the following weaknesses and cannot be served as the primary investment selection criteria:

1. It is not a true rate of return of the project.
2. It is based on accounting net income and book values, instead of the actual cash flows and market values.
3. Time value of money is ignored.
4. It uses an arbitrary benchmark cut-off rate.

Payback period has the following weaknesses and cannot be served as the primary investment selection criteria:

1. Time value of money is ignored.
2. It uses an arbitrary benchmark cut-off period of time to get back the initial investment capital.
3. It ignores cash flows after the cut-off date of the payback period.
4. It biases against long-term projects, such as research and development.

Answer 4(a)

$$\begin{aligned}\text{Net income} &= \text{Sales} - \text{Expenses} - \text{Taxes} \\ &= \text{HK\$4,507,000} - (\text{HK\$1,352,100} + \text{HK\$1,417,000} + \text{HK\$611,000} + \text{HK\$294,000}) \\ &\quad - \text{HK\$166,580} \\ &= \text{HK\$666,320}\end{aligned}$$

Answer 4(b)(i)

- 1) Net profit margin = $\text{HK\$666,320} / \text{HK\$4,507,000} = 14.78\%$
 - 2) Total asset turnover = $\text{HK\$4,507,000} / \text{HK\$3,888,800} = 1.16$
 - 3) Equity multiplier = $\text{HK\$3,888,800} / \text{HK\$2,013,200} = 1.93$
- Return on equity ("ROE") = Net profit margin x Total asset turnover x Equity multiplier
= 33.09%

Answer 4(b)(ii)

The DuPont analysis of ROE enables a company to evaluate the financial performance in following ways:

The DuPont analysis breaks down the factors affecting ROE into three different aspects covering components both in the statement of profit or loss and statement of financial position.

The company can rectify any problems identified as reflected by the assessments of the three financial analysis, including (1) operating efficiency measured by the net profit margin, (2) asset use efficiency measured by the total asset turnover, and (3) financial leverage measured by the equity multiplier.

Answer 4(c)

$$\text{Retention rate} = b = (\text{HK\$666,320} - \text{HK\$220,000}) / \text{HK\$666,320} = 0.6698$$

$$\text{Return on asset ("ROA")} = \text{HK\$666,320} / \text{HK\$3,888,800} = 17.13\%$$

Internal growth rate:

$$g = \frac{\text{ROA} \times b}{1 - \text{ROA} \times b} = \frac{17.13\% \times 0.6698}{1 - 17.13\% \times 0.6698} = 12.96\%$$

Answer 4(d)

Sustainable growth rate should be higher than the internal growth rate because:

- By comparing the formulas for sustainable growth rate and internal growth rate, ROE is larger than ROA.
- For the sustainable growth rate, Little Ox will have to borrow additional debt to maintain a constant debt to equity ratio. The total assets will be increased by this extra debt financing in addition to the internally generated fund from retained earnings. As such, Little Ox can expand more rapidly than the internal growth rate.

* * * END OF EXAMINATION PAPER * * *